

Topic: Introduction to Computer Organization and Architecture

Q. Define Computer Organization and Computer Architecture. Clearly distinguish between the two concepts using suitable examples.

📌 Definition to Remember

Computer Architecture defines the logical structure and functional behavior of a computer system as seen by the programmer (the "what"). **Computer Organization** refers to the physical hardware components and their interconnections that implement the architecture (the "how").

1. Computer Architecture (The "What")

- Deals with the attributes of a system visible to a programmer.
- Defines the logical blueprint and functional requirements.
- **Key Components:** Instruction set, data types, addressing modes, number of bits used to represent data, and I/O mechanisms.
- **Changeability:** Usually remains the same across a family of processors so old software can run on new machines.

2. Computer Organization (The "How")

- Deals with the operational units and their interconnections.
- Hardware details transparent to the programmer.
- **Key Components:** Control signals, circuit design, adders, memory technology, interfaces between peripherals.
- **Changeability:** Changes frequently with advancing technology to improve speed and efficiency.

3. Key Differences

Feature	Computer Architecture	Computer Organization
Focus	Focuses on What the computer does	Focuses on How the computer does it
Visibility	Visible to the programmer	Transparent to the programmer (hardware level)
Components	Instruction sets, data formats, addressing	Circuits, adders, memory technology, control signals
Example	Does the system have a multiply instruction?	Is multiplication implemented via a special circuit or repeated addition?

4. Real-World Example

The **Intel x86 family** architecture has remained largely compatible from the original 8086 up to modern processors (allowing old code to run). However, its **organization** (cache structures, pipelining, physical chips,

clock speeds) has changed drastically over decades to improve performance.

★ **Must-Write Points (for 10 marks)**

1. Computer Architecture = The "What"; Computer Organization = The "How".
 2. **Architecture** involves attributes visible to the programmer (logical structure).
 3. Architecture components: Instruction sets, addressing modes, data types.
 4. **Organization** involves hardware units and interconnections.
 5. Organization components: Circuit design, memory technology, control signals.
 6. Architecture is stable across processor families (for software compatibility).
 7. Organization changes rapidly with technology (e.g., adding caches, better pipelines).
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⚡ **Quick Recall**

Architecture (What it does, programmer visible, Instruction sets, stable) vs Organization (How it does it, hardware/circuits, changes with tech)